

MARICO SALESIQ — SYSTEM DOCUMENTATION

Technical Specification: Architecture, Data Processing Logic & Model Assumptions

System: Marico SalesIQ Intelligence Platform	Target Market: KSA Modern Trade (8 Retailers)	Classification: Technical Specification
Data Scope: 14,416 Records (12 Months)	Categories: Hair Oils, Creams, Shampoo	Brands: Marico, Parachute (+9 Competitors)

1. System Architecture & Modular Design

Marico SalesIQ follows a **modular, decoupled multi-tier architecture** that isolates data ingestion, econometric modeling, cognitive agent reasoning, and presentation layers:

Architectural Layer	Component / File	Core Functional Capabilities
1. Ingestion & Pipeline Layer	sales_agent/data_pipeline.py SalesDataPipeline	Sanitizes Excel inputs, derives per-unit sell-in/sell-out pricing, calculates macro/micro non-promo baselines, and constructs the unified df_merged master dataset (14,416 rows).
2. Econometric Analytics Layer	sales_agent/ • forecaster.py • rpi_engine.py • activation_optimizer.py • promotion_roi.py • distribution_engine.py	Five specialized mathematical engines executing ARDL Ridge forecasting, log-log price elasticity regressions, CDI/BDI/FSI fair share opportunity matrices, BTL trade spend ROI models, and ACV weighted distribution elasticity models.
3. AI Intelligence & Orchestration	sales_agent/agent.py SalesIntelligenceAgent	Central natural language query router and decision engine that maps executive questions to underlying econometric models and synthesizes prescriptive textual and tabular answers.
4. Presentation & Delivery Layer	• app.py (Streamlit Web App) • run_analysis.py (CLI Batch) • generate_pdf_report.py	Interactive 8-tab executive cockpit featuring Plotly dark visualizations, scenario sliders, chat assistant, and one-click PDF generation.

2. Data Processing Logic & Mathematical Formulations

The ingestion and feature engineering pipeline processes raw commercial feeds through a strict 6-stage lifecycle:

- **1. Data Sanitization:** Corrects vendor naming inconsistencies (e.g. standardizes TRESemm to TRESemme) and trims string whitespace across categorical columns.
- **2. Unit Price Derivation:** Derives real transaction unit price before merging: $Sell_in_Price = Sell_in_Value / Sell_in_Units$ and $Sell_out_Price = Sell_out_Value / Sell_out_Units$.
- **3. Multi-Key Harmonization:** Merges Sell-In and EPOS Sell-Out using composite key [Month, Chain, Category, Brand, SKU] across 14,416 rows with 100% record retention.
- **4. Pipeline Inventory Dynamics:** Computes monthly channel stock shifts via $Inventory_Delta = Sell_in_Units - Sell_out_Units$ and evaluates cumulative stock cover days.
- **5. Non-Promotional Baseline Decomposition:** For each SKU-Chain, average volume during Scheme == 'No Promotion' forms baseline volume. Fallback hierarchical macro-baselines prevent zero division. Incremental volume is computed as: $Incremental_Units = \max(0, Sell_out_Units - Baseline_Units)$.
- **6. Relative Price Index (RPI):** Benchmarks brand price against the chain-category average price: $RPI = (Brand_Price / Category_Avg_Price) * 100$.

Analytical Metric	Mathematical Formulation	Commercial Purpose
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Fair Share Index (FSI)	$FSI = (Chain_Market_Share / Benchmark_Share) * 100$	Identifies under-indexed accounts (<98 FSI) requiring ATL marketing and Endcap visibility.
Opportunity Gap (SAR)	$Gap = \max(0, Benchmark_Share - Chain_Share) * Chain_Cat_Turnover$	Quantifies uncaptured revenue in SAR if the brand achieves fair-share portfolio share.
Price Elasticity (PED)	$\ln(Units) = \alpha + \beta * \ln(Price) + \varepsilon \text{ (PED} = \beta\text{)}$	Determines if category demand is Inelastic (Hair Oils: -0.75) or Elastic (Shampoo: -1.10).
Trade Spend ROI (%)	$ROI = [(Incremental_Rev * 45\% - Trade_Discount) / Trade_Discount] * 100$	Evaluates net gross profit return per SAR of BTL promotional discount spend.
Weighted Distribution (WD) Elasticity	$\Delta Market_Share = 0.085 * \Delta WD\%$	Rule of thumb: Every +1.0% increase in Weighted Distribution yields +0.085% gain in market share.

3. Key Model Assumptions & Commercial Hypotheses

To ensure statistical validity and commercial realism, the models incorporate the following standard parameters:

Parameter / Domain	Standard Value / Assumption	Commercial Rationale & Industry Alignment
Gross Margin Benchmark	45.0% of Incremental Revenue	Standard FMCG personal care gross margin threshold after deducting COGS and variable costs.
Replenishment Lead-Lag	0 to 1 Month Pull-Through (r = 0.84 – 0.99)	Verified via cross-correlation; distributor sell-in synchronizes with retail EPOS offtakes within 0–30 days.
Price Elasticity Kink	Asymmetric demand penalty when RPI > 100	FMCG consumers exhibit sharp drop-off when pricing exceeds benchmark competitor levels.
Trade Scheme Discount Mechanics	15% Price Off: 15% discount - BOGO: 50% discount - Flat 2 SAR: 2.0 SAR/unit 25% Extra Vol: 20% effective cost	Models true brand and retailer funding requirements for BTL trade promotion schemes.
Pipeline Health Thresholds	- High Stockout Risk: < 15 Days - Optimal Buffer: 15 – 90 Days - Excess Overhang: > 90 Days	Buffers distributor stock to support May–July summer demand surge without creating dead inventory.
Portfolio Boundary	Marico Portfolio = Marico & Parachute	Harmonizes internal brand portfolio against 9 external multinational and regional competitors.

4. How to Generate & Send as PDF Format

Delivery Options for Evaluators / Stakeholders:

- Direct Technical PDF:** Run `python generate_documentation_pdf.py` in the terminal to generate `Marico_SalesIQ_Technical_Documentation.pdf`.
- Commercial Strategy PDF:** Run `python generate_pdf_report.py` to generate `Marico_Sales_Intelligence_Final_Report.pdf`.
- Interactive Web App Export:** Run `streamlit run app.py`, open **Tab 8 (Final Executive Report)**, and click **Download Official Report (PDF)**.
- Distribution:** Both generated PDF files are self-contained, publication-ready vector documents that can be attached directly to email or uploaded to your submission portal.